

Dataset: [Titanic_Dataset](#)

Assignment Questions (Visualization Based)

Q1. Create a countplot to show the number of passengers by gender.

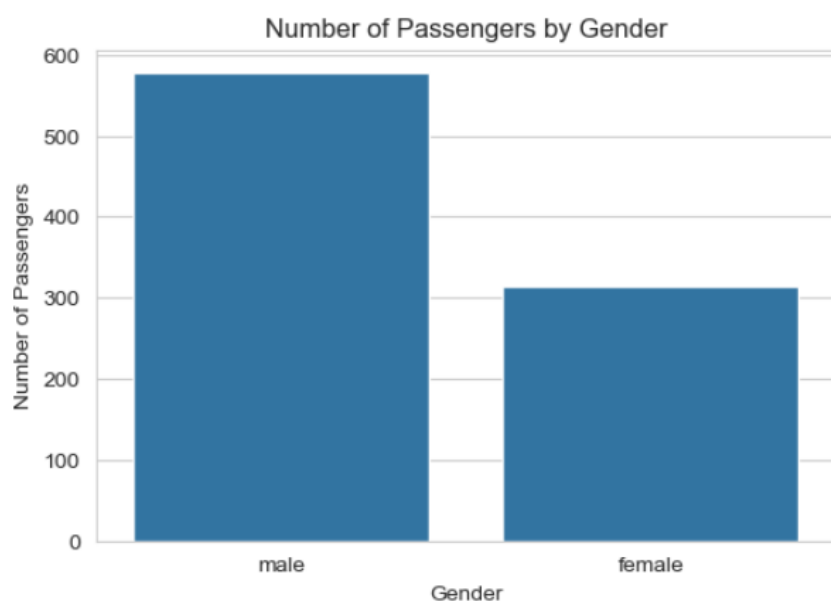
Ans:-

```
import matplotlib.pyplot as plt
import seaborn as sns

# Countplot of passengers by gender
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', data=df)

plt.title('Number of Passengers by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')

plt.show()
```



Interpretation

- The countplot displays the distribution of passengers based on gender.
- The number of male passengers is significantly higher than the number of female passengers.
- This indicates that men made up the majority of travelers on the Titanic.

Insight

The Titanic dataset contains more male passengers than female passengers, showing a gender imbalance among the travelers aboard the ship. This difference may influence survival analysis and other patterns in the dataset.

Q2. Create a bar chart to show the number of passengers in each passenger class using different colors.

Ans:-

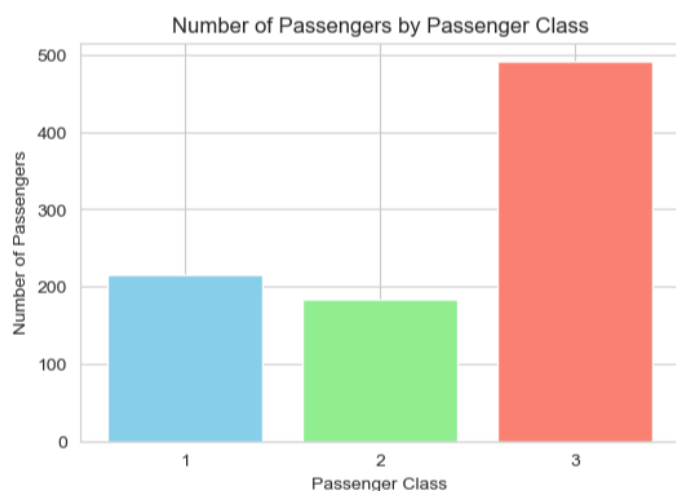
```
import matplotlib.pyplot as plt
import seaborn as sns

# Count passengers in each class
class_counts = df['Pclass'].value_counts().sort_index()

# Create bar chart
plt.figure(figsize=(6,4))
plt.bar(class_counts.index,
        class_counts.values,
        color=['skyblue', 'lightgreen', 'salmon'])

plt.title('Number of Passengers by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.xticks([1, 2, 3])

plt.show()
```



Interpretation

- The bar chart shows the distribution of passengers across the three passenger classes.
- **Third Class (Pclass 3)** has the highest number of passengers.
- **First Class (Pclass 1)** has fewer passengers than Third Class but more than Second Class.
- **Second Class (Pclass 2)** has the smallest number of passengers among the three classes.

Insight

Most passengers traveled in Third Class, indicating that the Titanic carried a large number of lower-fare travelers. Passenger class may have played an important role in survival rates and overall passenger demographics, making it an important variable for further analysis.

Q3. Create a histogram to show the distribution of passenger age.

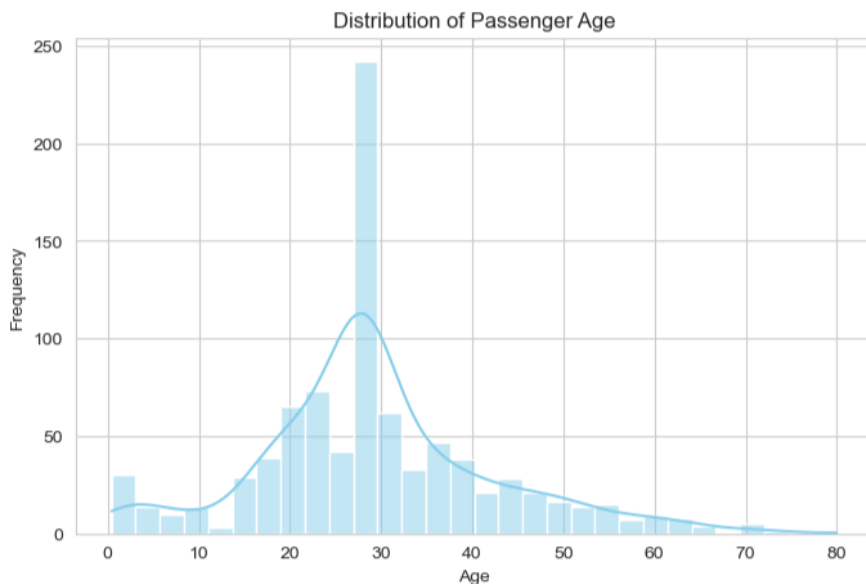
Ans:-

```
import matplotlib.pyplot as plt
import seaborn as sns

# Histogram of passenger age
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True, color='skyblue')

plt.title('Distribution of Passenger Age')
plt.xlabel('Age')
plt.ylabel('Frequency')

plt.show()
```



Interpretation

- The histogram displays how passengers are distributed across different age groups.
- Most passengers were young adults between 20 and 40 years old.
- There are relatively fewer elderly passengers compared to younger adults.
- The KDE (Kernel Density Estimation) curve helps visualize the overall age distribution pattern.
- The distribution is slightly right-skewed, indicating that younger passengers were more common than older passengers.

Insight

The majority of Titanic passengers were adults in the 20–40 age range. Children and elderly passengers formed a smaller proportion of the total passenger population. Age may be an important factor when analyzing survival patterns on the Titanic.

Q4. Create a boxplot to detect outliers in the fare column.

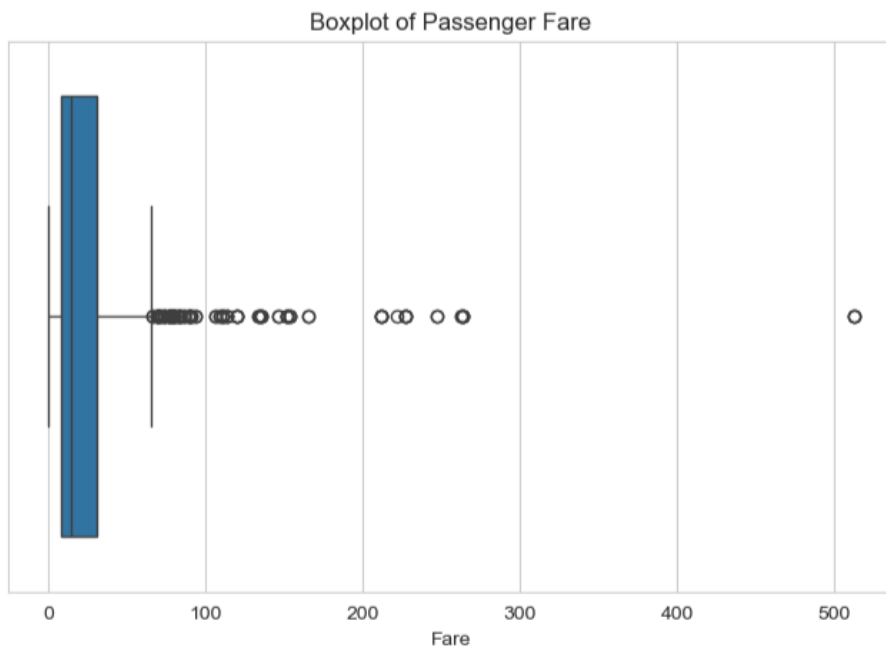
Ans:-

```
import matplotlib.pyplot as plt
import seaborn as sns

# Boxplot for Fare
plt.figure(figsize=(8,5))
sns.boxplot(x=df['Fare'])

plt.title('Boxplot of Passenger Fare')
plt.xlabel('Fare')

plt.show()
```



Interpretation

- The boxplot displays the distribution of passenger fares and helps identify outliers.
- The box represents the interquartile range (IQR), containing the middle 50% of fare values.
- The line inside the box indicates the median fare.
- Points lying beyond the whiskers are considered potential outliers.
- Several passengers paid exceptionally high fares compared to the majority, resulting in numerous outliers on the higher end of the fare distribution.

Insight

The Fare column contains significant outliers, indicating that a small number of passengers paid much higher ticket prices than most others. This is expected because First-Class passengers often paid substantially more than passengers in Second or Third Class. The fare distribution is therefore highly right-skewed.

Q5. Create a scatterplot to

- show the relationship between age and fare.
- analyze the relationship between family size (SibSp + Parch) and fare.

Ans:-

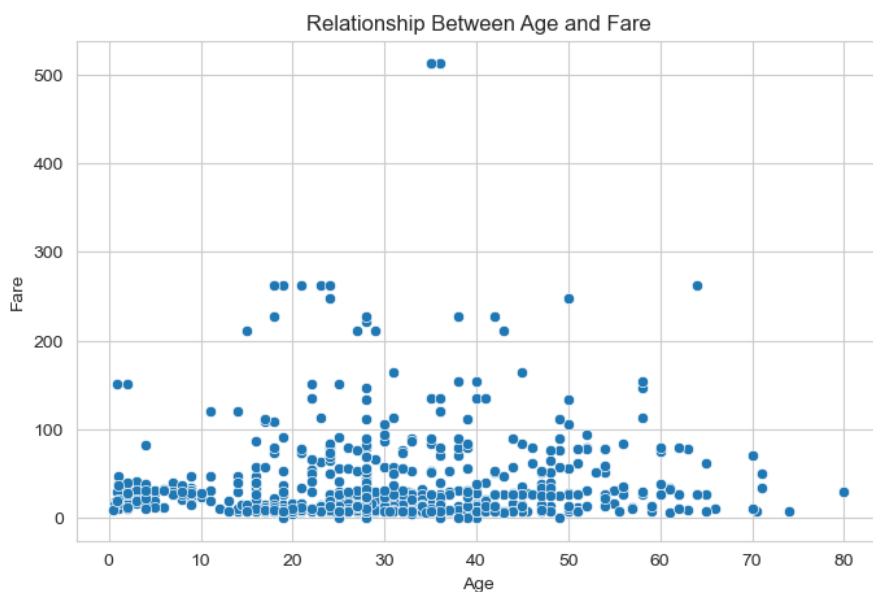
(a). Create a Scatterplot to Show the Relationship Between Age and Fare

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,5))
sns.scatterplot(x='Age', y='Fare', data=df)

plt.title('Relationship Between Age and Fare')
plt.xlabel('Age')
plt.ylabel('Fare')

plt.show()
```



Interpretation

- The scatterplot shows the relationship between passenger age and ticket fare.
- Most passengers paid lower fares regardless of age.
- There is no strong linear relationship between age and fare.
- Some passengers paid exceptionally high fares, creating visible outliers.

Insight

Age does not appear to have a significant impact on the fare paid. Ticket price was likely influenced more by passenger class than by age.

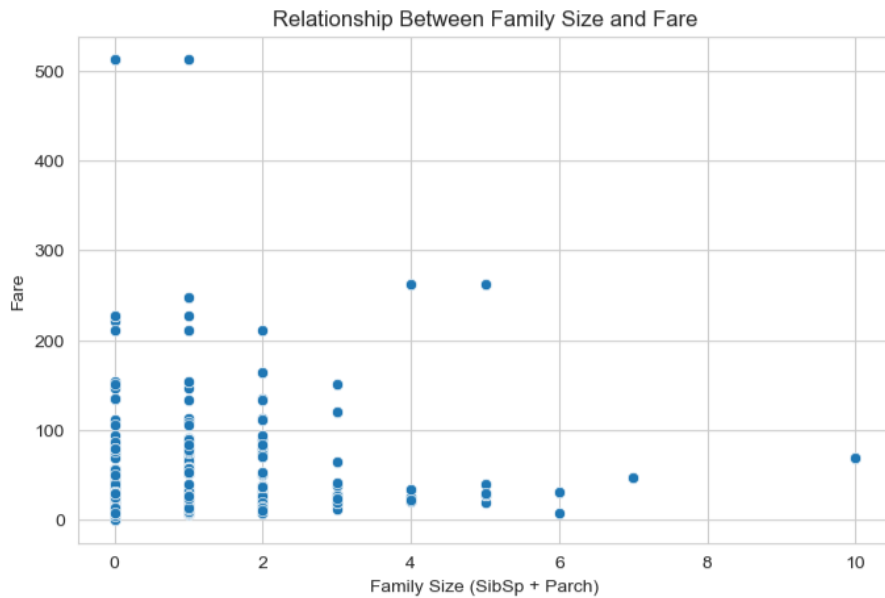
(b). Create a Scatterplot to Analyze the Relationship Between Family Size and Fare

```
# Create Family Size column
df['FamilySize'] = df['SibSp'] + df['Parch']

# Scatterplot
plt.figure(figsize=(8,5))
sns.scatterplot(x='FamilySize', y='Fare', data=df)

plt.title('Relationship Between Family Size and Fare')
plt.xlabel('Family Size (SibSp + Parch)')
plt.ylabel('Fare')

plt.show()
```



Interpretation

- The scatterplot shows how ticket fare varies with family size.
- Most passengers traveled alone or with small families.
- Higher fares are generally observed among passengers with small to medium-sized families.
- There is no strong positive or negative correlation between family size and fare.

Insight

Family size does not have a clear direct relationship with fare. While some passengers with larger families paid higher fares, fare variation appears to be influenced more by passenger class and ticket type than by family size alone.

Q6. Create a countplot to show survival count by gender.

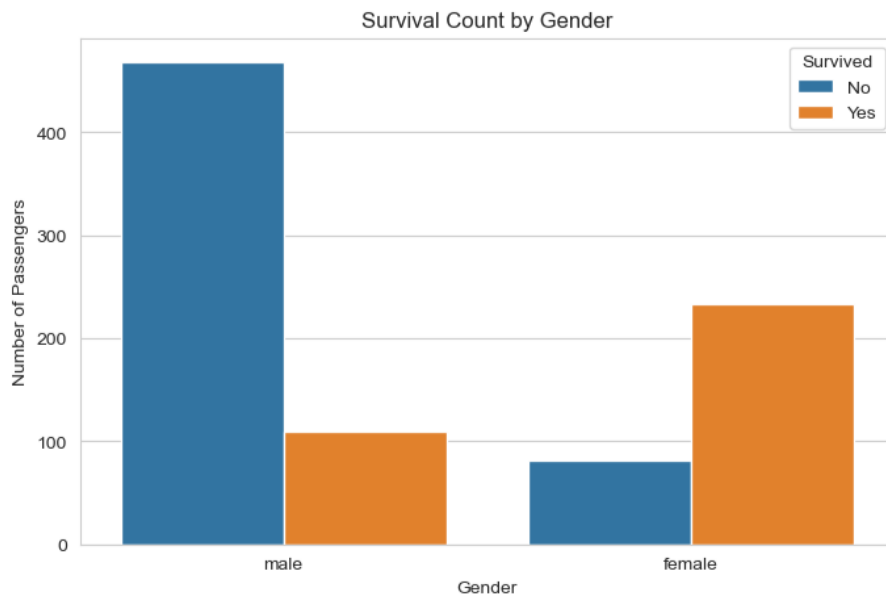
Ans:-

```
import matplotlib.pyplot as plt
import seaborn as sns

# Countplot of survival by gender
plt.figure(figsize=(8,5))
sns.countplot(x='Sex', hue='Survived', data=df)

plt.title('Survival Count by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])

plt.show()
```



Interpretation

- The countplot compares survival counts for male and female passengers.
- Among female passengers, the number of survivors is significantly higher than the number who did not survive.
- Among male passengers, the number of non-survivors is much higher than the number of survivors.
- This indicates that gender played a major role in survival during the Titanic disaster.

Insight

Female passengers had a much higher survival rate than male passengers. This supports the historical "women and children first" evacuation policy followed during the Titanic disaster, which likely contributed to the higher number of female survivors.

Q7. Create a bar chart to show survival by passenger class.

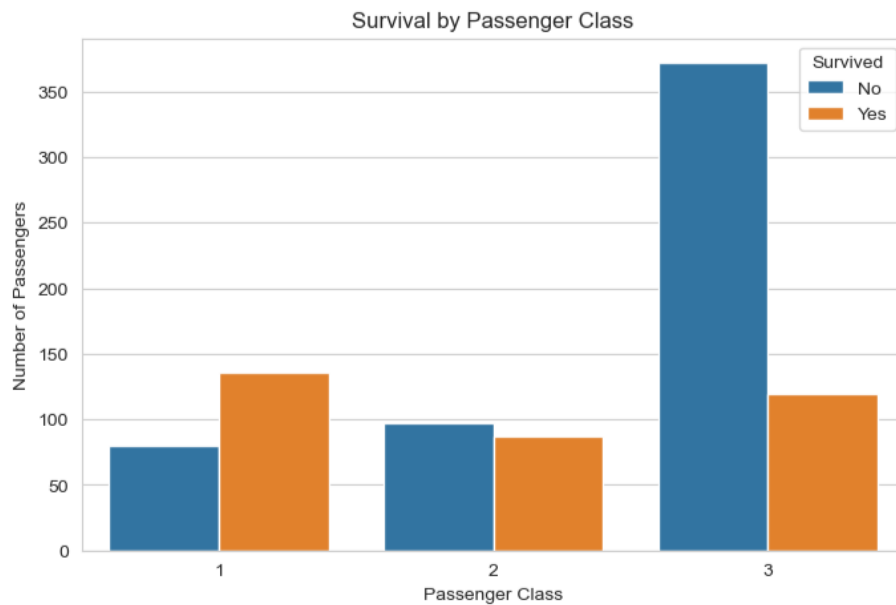
Ans:-

```
import matplotlib.pyplot as plt
import seaborn as sns

# Survival by passenger class
plt.figure(figsize=(8,5))
sns.countplot(x='Pclass', hue='Survived', data=df)

plt.title('Survival by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.legend(title='Survived', labels=['No', 'Yes'])

plt.show()
```



Interpretation

- The bar chart compares the number of survivors and non-survivors across passenger classes.
- First-Class passengers had a relatively high number of survivors.
- Third-Class passengers experienced the highest number of deaths.
- Second-Class passengers had survival counts between those of First and Third Class.

Insight

Passenger class had a significant impact on survival. First-Class passengers had a better chance of survival, while Third-Class passengers faced the highest mortality rate. This suggests that socioeconomic status and cabin location may have influenced access to lifeboats and evacuation opportunities during the disaster.

Q8. Create a heatmap to show correlation between numerical variables.

Ans:-

```
import matplotlib.pyplot as plt
import seaborn as sns

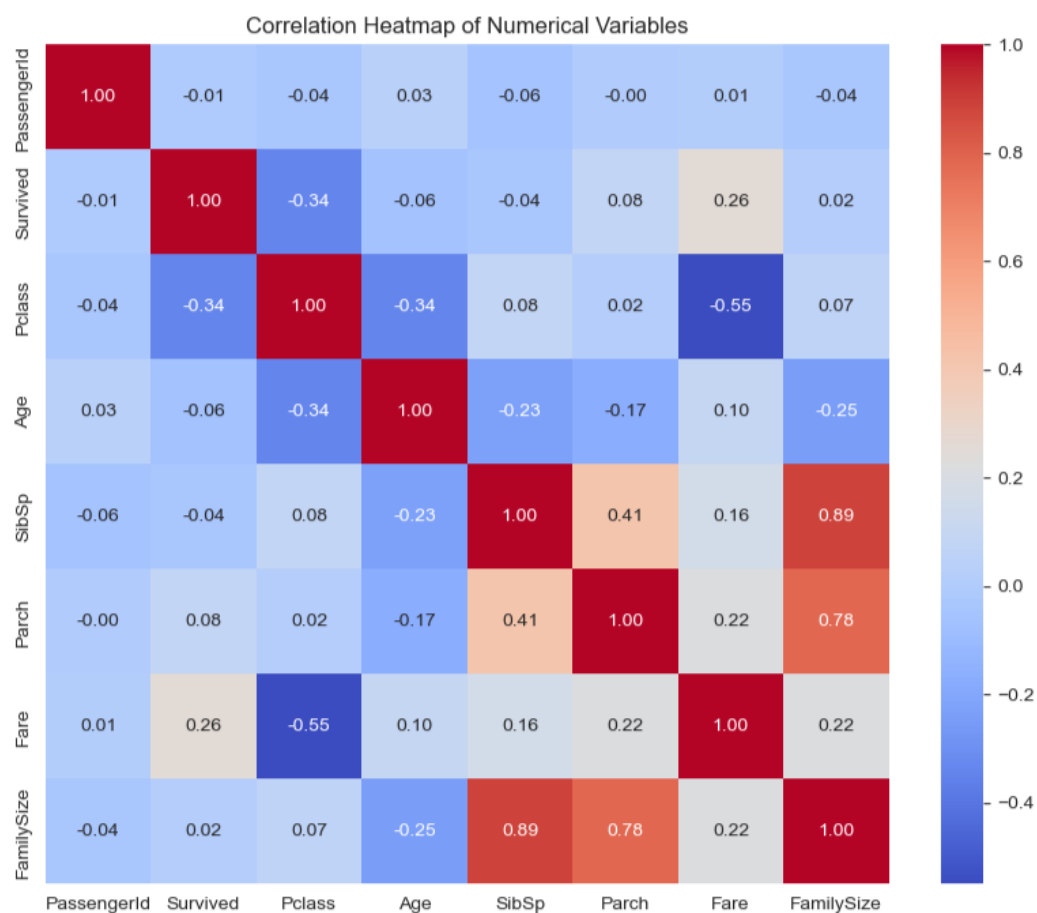
# Select numerical columns
numeric_df = df.select_dtypes(include=['int64', 'float64'])

# Compute correlation matrix
correlation_matrix = numeric_df.corr()

# Create heatmap
plt.figure(figsize=(10,8))
sns.heatmap(correlation_matrix,
            annot=True,
            cmap='coolwarm',
            fmt='.2f')

plt.title('Correlation Heatmap of Numerical Variables')

plt.show()
```



Interpretation

- The heatmap displays the correlation coefficients between numerical variables.
- Correlation values range from **-1 to +1**:
 - **+1** indicates a strong positive correlation.
 - **-1** indicates a strong negative correlation.

- **0** indicates little or no correlation.
- Variables with darker colors represent stronger relationships.
- The Survived variable may show noticeable relationships with variables such as Pclass, Fare, and Age.

Insight

The heatmap helps identify which numerical features are related to each other. In the Titanic dataset, passenger class and fare often show a strong relationship, while survival tends to be associated with factors such as passenger class and fare. This visualization is useful for understanding feature relationships before building predictive models.

Q9. Identify which passenger class paid the highest fare on average.

Ans:-

```
# Calculate average fare by passenger class
average_fare = df.groupby('Pclass')['Fare'].mean()

print("Average Fare by Passenger Class:")
print(average_fare)

# Bar chart
import matplotlib.pyplot as plt

average_fare.plot(kind='bar', color=['skyblue', 'lightgreen', 'salmon'])

plt.title('Average Fare by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Average Fare')
plt.xticks(rotation=0)

plt.show()
```

Average Fare by Passenger Class:

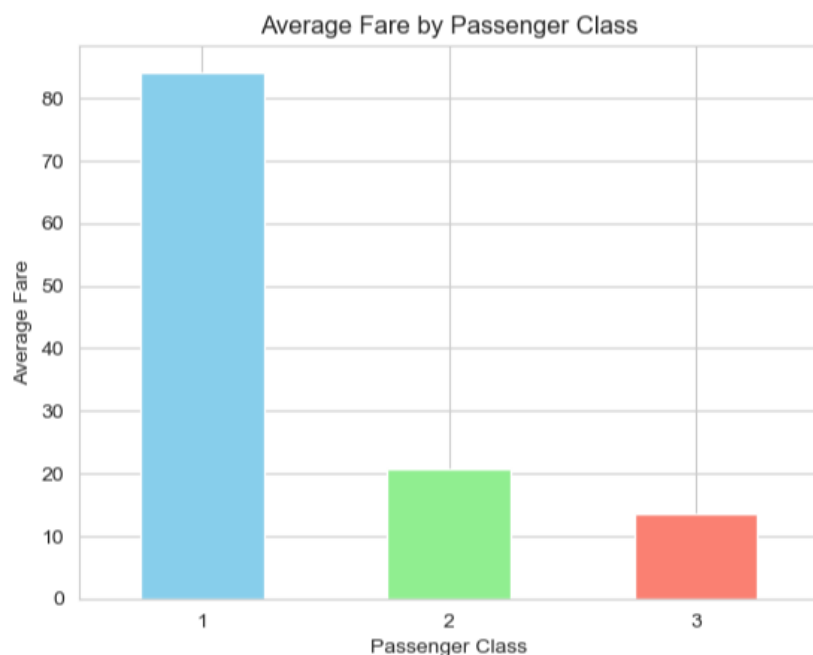
Pclass

1 84.154687

2 20.662183

3 13.675550

Name: Fare, dtype: float64



Interpretation

- The code calculates the average fare paid by passengers in each class.
- The bar chart visually compares the average fare across First, Second, and Third Class passengers.
- Typically, First-Class passengers paid the highest average fare, followed by Second Class and then Third Class.

Answer

Passenger Class 1 (First Class) paid the highest fare on average among all passenger classes.

Insight

The average fare decreases as passenger class moves from First Class to Third Class. This indicates that First-Class passengers enjoyed premium accommodations and services, which were reflected in the higher ticket prices. This difference in fare may also be associated with the higher survival rates observed among First-Class passengers.

Q10. Write at least 5 insights from the dataset based on your visualizations.

Ans:-

Insights

1. Male passengers outnumbered female passengers

- The gender countplot showed that there were significantly more male passengers than female passengers on board the Titanic.

2. Most passengers traveled in Third Class

- The passenger class bar chart revealed that Third Class (Pclass 3) had the highest number of passengers, indicating that most travelers belonged to the lower fare category.

3. The majority of passengers were young adults

- The age histogram showed that most passengers were between 20 and 40 years old, with fewer children and elderly passengers.

4. Female passengers had a higher survival rate

- The survival-by-gender countplot demonstrated that a much larger proportion of women survived compared to men, supporting the "women and children first" evacuation policy.

5. Passenger class strongly influenced survival

- The survival-by-class chart indicated that First-Class passengers had higher survival rates, while Third-Class passengers experienced the highest number of deaths.